

RUOQIANG, China

WMO: 517770

Lat: 39.0239N

Lon: 88.1837E

Elev: 889

StdP: 91.09

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-14.2	-23.0	0.5	-9.1	-21.2	0.6	-7.4	6.6	-3.9	5.6	-5.4	1.4	250	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	38.3	19.3	36.9	18.8	35.5	18.4	21.9	32.8	20.8	31.9	19.9	31.7	2.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	15.1	26.6	17.3	13.8	25.6	16.1	12.8	25.2	68.9	32.6	64.5	31.9	61.3	31.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.2	6.9	5.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-17.8	41.2	2.0	1.2	-19.3	42.1	-20.5	42.8	-21.6	43.5	-23.1	44.3
			-18.3	25.2	1.9	1.5	-19.7	26.2	-20.8	27.1	-21.8	28.0	-23.2	29.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	-6.8	-0.2	8.7	16.4	21.5	26.1	28.0	26.2	20.8	12.6	3.8	-4.7	(6)
(7)		DBStd	12.41	3.18	3.95	4.51	4.27	3.36	2.65	2.21	2.34	2.91	3.59	3.94	3.36	(7)
(8)		HDD10.0	1551	520	287	81	6	0	0	0	0	15	188	456		(8)
(9)		HDD18.3	3032	778	520	301	83	11	0	0	10	179	436	714		(9)
(10)		CDD10.0	2564	0	0	39	198	357	484	558	503	325	97	2	0	(10)
(11)		CDD18.3	1003	0	0	1	26	110	234	300	245	84	2	0	0	(11)
(12)		CDH23.3	13227	0	0	23	460	1496	3019	3990	3084	1054	99	0	0	(12)
(13)		CDH26.7	6936	0	0	2	145	674	1656	2326	1688	426	19	0	0	(13)
(14)	Wind	WSAvg	2.2	1.8	2.2	2.6	2.9	2.8	2.6	2.3	2.2	2.1	1.9	1.8	1.7	(14)
(15)	Precipitation	PrecAvg	26	1	1	1	1	2	5	9	3	1	0	1	1	(15)
(16)		PrecMax	107	7	15	8	12	24	26	75	25	15	11	10	7	(16)
(17)		PrecMin	5	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	18	2	2	2	2	4	7	12	5	3	2	2	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	5.3	14.0	26.1	33.4	35.9	39.2	40.5	39.0	35.0	29.2	19.4	7.0		(19)
(20)		MCWB	-0.7	4.2	10.0	14.0	16.5	19.4	20.5	20.0	17.9	13.7	7.7	1.1		(20)
(21)		DB	2.5	11.1	23.0	30.2	34.1	37.4	38.8	37.3	32.9	26.1	16.4	4.5		(21)
(22)		MCWB	-2.3	2.8	8.7	12.6	15.7	18.3	19.7	19.3	16.9	12.5	6.4	-0.6		(22)
(23)		DB	0.5	9.2	20.8	28.1	32.5	35.9	37.4	35.9	31.1	23.6	13.9	2.5		(23)
(24)		MCWB	-3.5	1.7	7.8	11.8	15.1	17.9	19.2	18.9	16.4	11.2	5.3	-1.8		(24)
(25)		DB	-1.2	7.1	18.1	26.0	30.7	34.3	35.8	34.3	29.2	21.3	11.3	0.7		(25)
(26)		MCWB	-4.7	0.4	6.4	10.9	14.4	17.5	19.0	18.6	15.5	10.1	4.0	-3.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	-0.4	4.5	10.2	14.5	18.6	23.3	23.6	23.7	20.4	14.5	7.8	1.4		(27)
(28)		MCDWB	4.9	13.4	24.9	31.6	31.4	32.9	35.0	32.9	29.3	25.7	18.4	6.2		(28)
(29)		WB	-2.1	3.0	8.9	13.1	16.8	20.7	21.8	21.8	18.7	13.0	6.5	-0.3		(29)
(30)		MCDWB	2.1	10.6	22.4	28.9	31.2	31.9	33.7	31.8	28.6	24.6	15.8	4.0		(30)
(31)		WB	-3.4	1.8	7.8	12.2	15.8	19.4	20.7	20.7	17.5	11.8	5.5	-1.7		(31)
(32)		MCDWB	0.2	8.8	20.2	26.9	30.3	31.5	32.8	31.2	28.3	22.5	13.6	2.2		(32)
(33)		WB	-4.6	0.6	6.6	11.2	14.9	18.5	20.0	19.8	16.4	10.5	4.1	-2.8		(33)
(34)		MCDWB	-1.4	6.8	17.8	25.2	29.2	30.8	32.3	30.9	27.1	20.2	11.1	0.5		(34)
(35)	Mean Daily Temperature Range	MDBR	11.1	12.9	15.2	15.5	15.8	14.6	14.6	15.5	16.9	17.4	13.4	10.4		(35)
(36)		5% DB	12.7	14.7	18.5	18.7	19.0	18.3	18.6	19.0	19.8	20.0	16.8	11.9		(36)
(37)		MCWBR	8.9	8.7	9.0	7.7	7.5	6.7	6.5	7.3	8.2	8.7	8.9	7.8		(37)
(38)		MCDBR	12.4	14.3	17.7	16.8	16.9	14.2	14.7	15.7	17.7	19.2	16.5	11.5		(38)
(39)		MCWBR	8.8	8.5	8.7	7.2	7.4	6.5	6.5	7.5	8.2	8.3	8.8	7.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.391	0.471	0.568	0.662	0.681	0.703	0.685	0.659	0.620	0.510	0.434	0.394		(40)
(41)		taud	1.925	1.752	1.566	1.426	1.442	1.458	1.516	1.542	1.575	1.712	1.848	1.917		(41)
(42)		Ebn at Noon	726	710	683	653	655	641	648	648	635	661	665	683		(42)
(43)		Edh at Noon	137	187	251	306	307	302	283	269	246	192	147	128		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.76	3.69	5.04	6.05	6.84	6.93	6.67	6.22	5.57	4.48	3.16	2.44		(44)
(45)		RadStd	0.20	0.19	0.24	0.36	0.31	0.40	0.38	0.34	0.22	0.13	0.17	0.17		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page